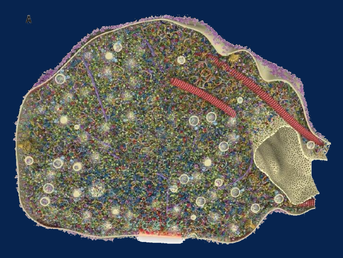

Ten Tips for Protein Property Prediction: Common Challenges and Best Practices

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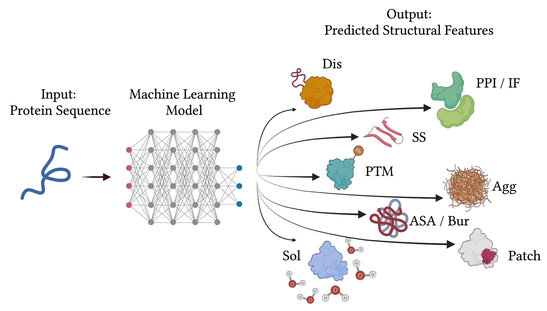
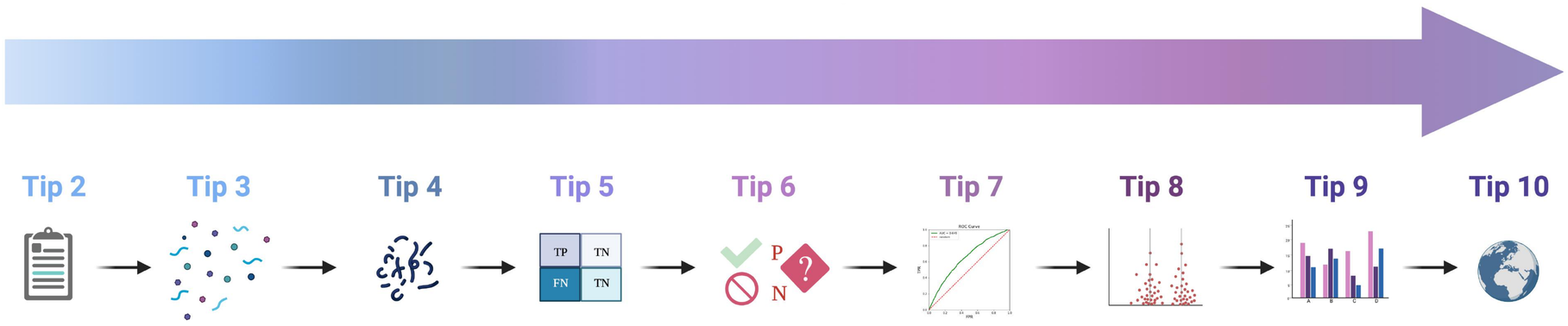
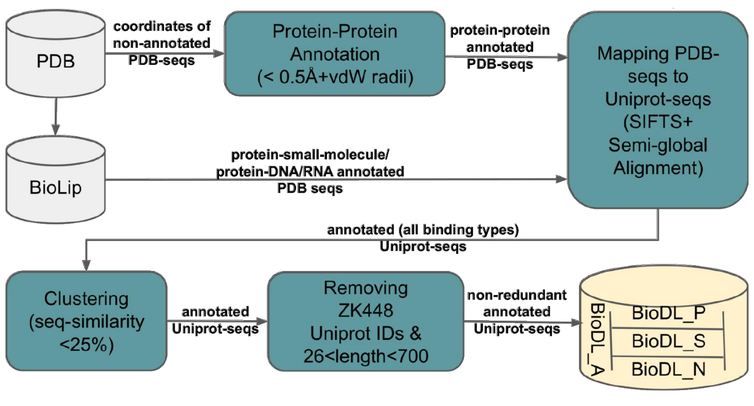
1. Your biological question



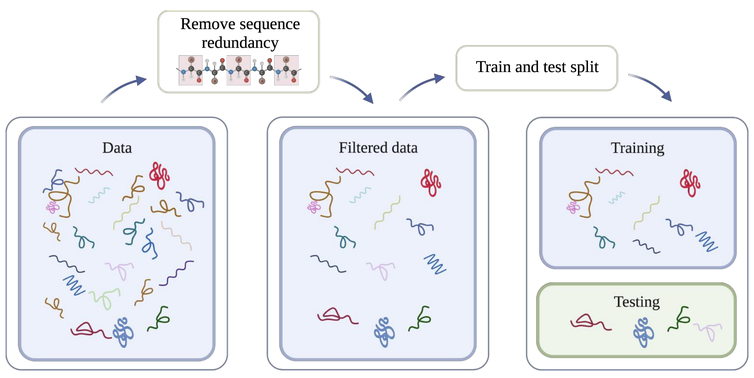


2. Describe & provide your datasets



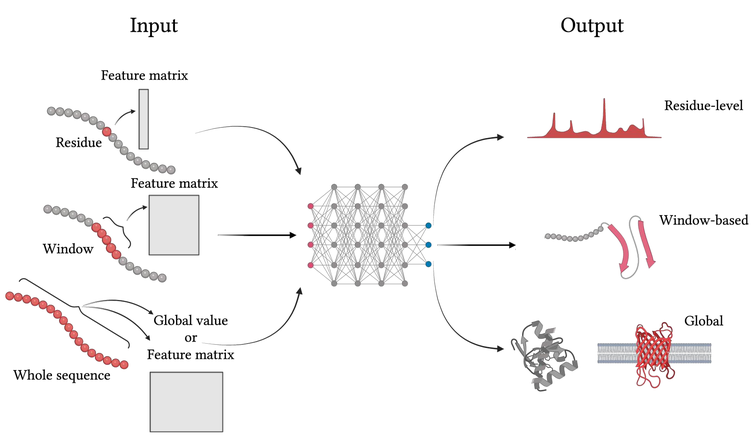
Be clear about how your dataset was generated. Where does the source data come from? How was it processed? Make your final dataset available, so others can benefit and build on your work.

3. Remove sequence redundancy



Homology between the training and test sets will skew the evaluation of your method performance. Also, overrepresented protein families in your training set will bias your method towards those types of proteins. Both problems are alleviated by filtering on sequence redundancy, e.g. at the level of 25% sequence identity (or even 15%).

4. Specify input and output



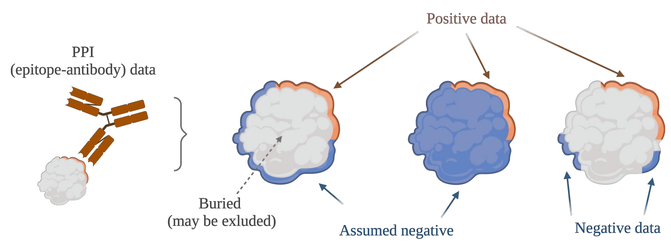
The input (features) of your model may be at the residue level, or sequence windows, or the whole sequence, and may also include global features such as protein length. Likewise, the output (predictions) may be at residue, window or global level. Make sure to define explicitly how your model input and output are organised.

5. Evaluation metrics

Name	Train	#	Test	#	Spec	Prec	Sens	F1	MCC	AUC
SeRenDIP [1]	HHC train	407	HHC test	142	0.721	0.299	0.618	0.403	0.235	0.706
ensnet_hhc [2]	HHC train	407	HHC test	139	**0.869**	0.405	0.405	0.405	**0.274**	0.741
ensnet_p [2]	BioDL train	4392	BioDL test	228	**0.909**	0.339	0.339	0.339	0.249	0.755
ensnet_p [2]	BioDL train	4392	ZK448 test	336	0.862	0.397	0.397	0.397	0.259	0.718
PInet (Aug 50) [3]	DBD3 Unbound	60	DBD3 LOOCV	13		**0.518**	**0.745**	**0.611**		**0.775**
PInet (Aug 50) [3]	DBD5 Unbound	189	DBD5 LOOCV	36		0.492	0.723	0.586		0.753

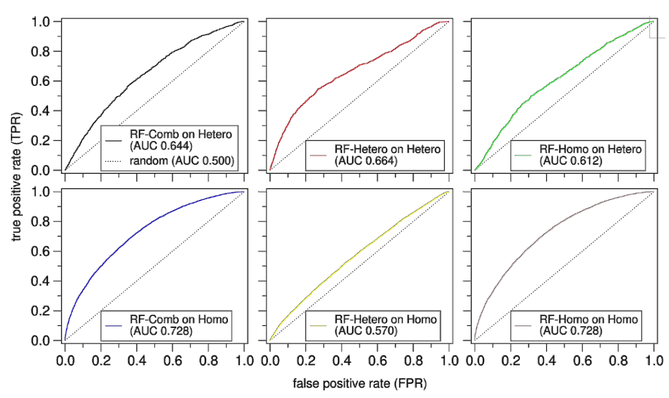
Not all (benchmark) papers always report many performance metrics. Since you're likely to use a script to calculate them, it is simple to calculate all metrics, such as specificity, precision, sensitivity, F1, MCC, AUC-ROC, all in one go. Put a relevant selection in the main text, and all others in the supplement.

6. Define positives & negatives



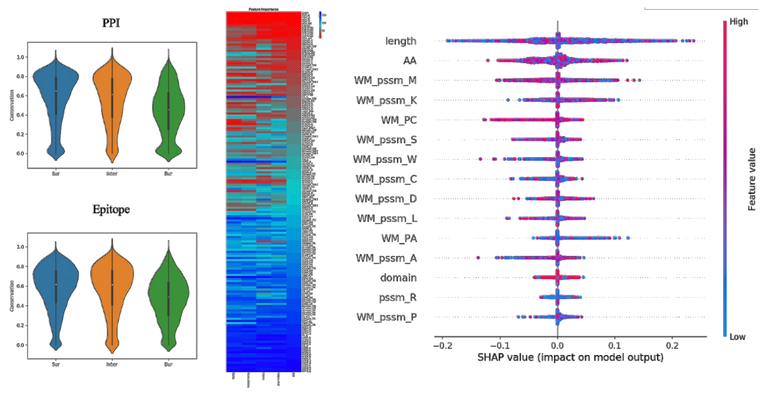
Depending on the type of experiment, most of your data will only indicate the positive class. For example, a crystal structure of a complex allows you to identify the binding region. But it does not exclude other proteins (or antibodies) to bind on other places. Mass-spectrometry data on epitope peptides may allow you to pinpoint also some negative regions, i.e. those that are clearly not being recognized by an antibody. Be clear on what your (experimental) data can tell you (and what not), and which limitations this implies for your training data, and your final trained model.

7. Label your results & data



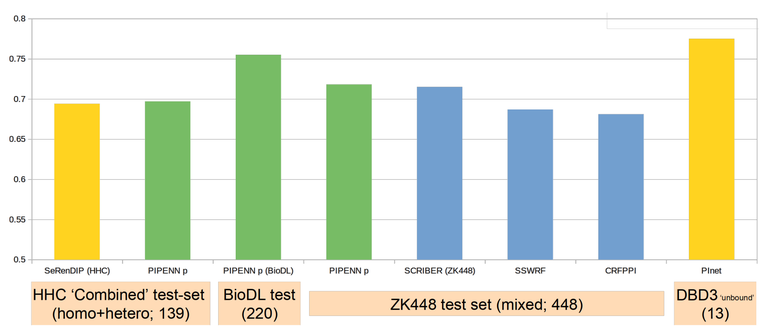
In most papers, you will be training and testing several models, perhaps stemming from different training sets and also applied to several (external) benchmark test sets. For all of the results that you present, make clear what the training set was, which model variation this is, and which test set was used. It doesn't hurt to be overly specific, e.g. something like: *"ROC plots showing the performance of Models RF-Comb, RF-Hetero, and RF-Homo, trained on the combined, heteromeric, and homomeric training sets, respectively, and tested on the heteromeric (Hetero) and homomeric (Homo) test sets"*

8. Interpret your models



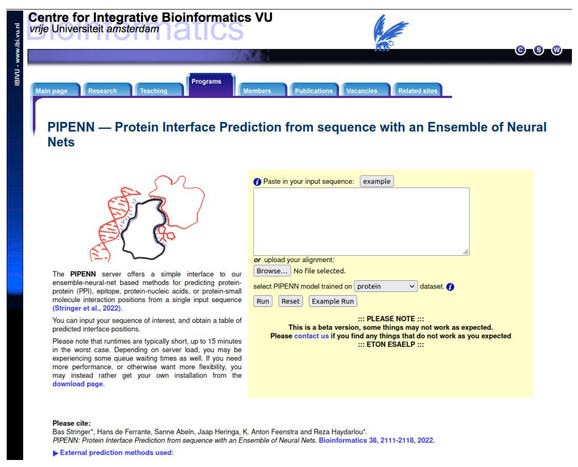
Your final trained model will only tell you the most likely property for each prediction, e.g. an antibody binding region in the sequence. To find out a bit about why this prediction was made this way, you will need to dive into your features, e.g. plotting distributions on various subsets (left). You can also look at feature importances, e.g. using gini importance (middle), or drill even deeper by exploring features' contribution to individual predictions, e.g. using SHAP values (right).

9. Fair comparison with others



For fair comparisons, you need to evaluate all methods on the same dataset (or datasets). Moreover, some methods use features that are harder to get, such as protein structure or derived data like DSSP, which gives them an edge over methods that only used cheaper and ubiquitous sequence data. Be clear which comparisons you make, and draw your conclusions accordingly.

10. Make your method available



Just as with your datasets, your methods will be valuable to other researchers. Sharing your code allows others to benefit from your progress, and be confident that this will swing back to you in due time. To enable non-experts to apply your method you may also want to provide it as a web server.

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